Gravity on the 7+1 spacetime and the coupling of dirac16complex to it

Einstein and Lovelock tensors, generalized frame matrices, the canonical spin connection with a vanishing total covariant derivative, the covariant Dirac equation of dirac16complex, and the primordial, modern and far-future epochs — sections 11–14 of the notebook 7+1.nb

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Abstract

This document puts the 16-component field dirac16complex of the 7+1 project into a gravitational field. The spacetime has ONE time coordinate x_0 and SEVEN space coordinates $x_1, \ldots, x_7$; the four auxiliary space-like coordinates $x_4, \ldots, x_7$ may expand or contract at their own rate. It defines and checks the curvature of a general metric, the Lovelock tensors of 7+1 spacetime (Einstein, Gauss–Bonnet, cubic; the quartic one vanishes identically in eight dimensions) and the Einstein–Lovelock field equations; the generalized frame matrices $\Gamma_\mu(X) = e^A{}_\mu(X)\,\Gamma_A$ built from the real 16×16 Dirac matrices; the canonical spin connection, for which the total covariant derivative of every frame matrix vanishes; the covariant Dirac equation, in which that spin connection couples gravity to dirac16complex; and the three cosmological epochs — primordial (near $x_0 = 0$), modern, far future — with the creation of dirac16complex pairs in each. The ideas follow the repository Dirac_claude, rebuilt for 7+1: Dirac_claude uses a 4+4 spacetime with four time-like coordinates and a warped metric, none of which is used here. Sections 11–14 of 7+1.nb contain 97 checks (97 True); the independent Python program tools/ref_gravity.py contains 99 (99 True).

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1. Results at a glance

Spacetime. Coordinates $X = (x_0, x_1, \ldots, x_7)$ with x_0 the time (the direction of the octonion unit $e_0 = 1$), x_1, x_2, x_3 ordinary space and $x_4, \ldots, x_7$ auxiliary space. The metric $g_{\mu\nu}(X)$ has signature $(+,-,-,-,-,-,-,-)$. The frame metric and the eight real Dirac matrices are

$$\eta = \mathrm{diag}(+1,-1,\ldots,-1), \qquad \Gamma_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \quad \Gamma_i = \begin{pmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{pmatrix}, \qquad \Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\eta_{AB}1_{16},$$

with the real antisymmetric 8×8 octonion matrices τ_i^O ($i = 1, \ldots, 7$, $(\tau_i^O)^2 = -1_8$, mutually anticommuting).

Lovelock tensors of order k, with the generalized Kronecker delta $\delta^{\mu_1\cdots\mu_n}_{\nu_1\cdots\nu_n} = \det[\delta^{\mu_i}_{\nu_j}]$:

$$E^{(k)\mu}{}_{\nu} = -\frac{1}{2^{k+1}}\,\delta^{\mu\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\nu\gamma_1\delta_1\cdots\gamma_k\delta_k}\,R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k},$$
$$E^{(0)} = -\tfrac{1}{2}\,\delta, \quad E^{(1)} = G \text{ (Einstein)}, \quad E^{(4)} \equiv 0 \text{ in } D = 8.$$

Einstein–Lovelock field equations ($\alpha_0 = 2\Lambda$, $\alpha_1 = 1$, $\kappa = 8\pi G_8$):

$$\sum_{k=0}^{3} \alpha_k\, E^{(k)\mu}{}_{\nu} = \kappa\, T^{\mu}{}_{\nu}.$$

Generalized frame matrices, canonical spin connection, total covariant derivative:

$$\Gamma_\mu(X) = e^A{}_\mu(X)\,\Gamma_A, \qquad \Gamma_\mu\Gamma_\nu + \Gamma_\nu\Gamma_\mu = 2g_{\mu\nu}(X)\,1_{16},$$
$$\omega_\mu{}^A{}_B = e^A{}_\nu\big(\partial_\mu e_B{}^\nu + \{{}^{\nu}_{\mu\lambda}\}\,e_B{}^\lambda\big), \qquad \Omega_\mu = \tfrac{1}{4}\,\omega_{\mu AB}\,\Gamma^A\Gamma^B,$$
$$\nabla_\mu\Gamma_\nu = \partial_\mu\Gamma_\nu - \{{}^{\lambda}_{\mu\nu}\}\Gamma_\lambda + [\Omega_\mu, \Gamma_\nu] = 0.$$

Covariant Dirac equation of dirac16complex ($D_\mu = \partial_\mu + \Omega_\mu$, $\bar\psi = \psi^\dagger\Gamma_0$):

$$\big(i\,\Gamma^\mu(X)\,D_\mu - m\big)\psi = 0, \qquad (\Gamma^\mu D_\mu)^2 = g^{\mu\nu}\nabla_\mu D_\nu - \tfrac{R}{4}\,1_{16}.$$

Three epochs of the cosmological metric $ds^2 = dx_0^2 - a^2(dx_1^2 + dx_2^2 + dx_3^2) - b^2(dx_4^2 + \cdots + dx_7^2)$: primordial Kasner solutions near $x_0 = 0$ (Einstein: $a = x_0^p$, $b = x_0^q$ with $p = (1 \pm 2\sqrt{2})/7$, $q = (2 \mp 3\sqrt{2})/14$; Gauss–Bonnet and cubic Lovelock exponents below), a modern epoch with a static auxiliary space ($b = b_0$, ordinary Friedmann cosmology with $G_4 = G_8/V_\perp$), and de Sitter far futures, in which dirac16complex pairs are created with the Fermi–Dirac distribution at the Gibbons–Hawking temperature, $|\beta|^2 = 1/(e^{2\pi M/H} + 1)$.

2. Conventions

- Units: $\hbar = c = 1$. A prime is d/dx_0.
- Indices: Greek $\mu, \nu, \lambda, \rho, \sigma, \alpha, \beta, \gamma, \delta \in \{0, \ldots, 7\}$ are coordinate indices; Latin capitals $A, B, C \in \{0, \ldots, 7\}$ are frame indices, raised and lowered with η; repeated indices are summed. In Mathematica all lists are 1-based: $g_{\mu\nu}$ is `g[[μ+1, ν+1]]`.
- Christoffel symbols are written $\{{}^{\lambda}_{\mu\nu}\}$ — never Γ, which in this project always denotes a Dirac matrix:

$$\{{}^{\lambda}_{\mu\nu}\} = \tfrac{1}{2}\,g^{\lambda\rho}\big(\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\big).$$

Curvature (the convention of Landau and Lifshitz):

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \{^{\rho}_{\sigma\nu}\} - \partial_\nu \{^{\rho}_{\sigma\mu}\} + \{^{\rho}_{\mu\lambda}\}\{^{\lambda}_{\sigma\nu}\} - \{^{\rho}_{\nu\lambda}\}\{^{\lambda}_{\sigma\mu}\},$$
$$R_{\sigma\nu} = R^{\rho}{}_{\sigma\rho\nu}, \qquad R = g^{\sigma\nu}R_{\sigma\nu}, \qquad G^{\mu}{}_{\nu} = R^{\mu}{}_{\nu} - \tfrac{1}{2}\delta^{\mu}_{\nu}R, \qquad R^{\gamma\delta}{}_{\alpha\beta} = g^{\delta\sigma}R^{\gamma}{}_{\sigma\alpha\beta}.$$

With these conventions and the signature $(+,-,\ldots,-)$ the field equations of general relativity read $G^{\mu}{}_{\nu} - \Lambda\delta^{\mu}_{\nu} = \kappa T^{\mu}{}_{\nu}$ with $T^{0}{}_{0} > 0$ the energy density, and a de Sitter space has $R < 0$.

3. Curvature of a 7+1 metric (section 11 of 7+1.nb)

The notebook function `curvature[g, simp]` returns, for an 8×8 metric in the coordinates X, the inverse metric, the Christoffel symbols, $R^{\rho}{}_{\sigma\mu\nu}$, $R^{\gamma\delta}{}_{\alpha\beta}$, the Ricci tensor, R and $G^{\mu}{}_{\nu}$. Three test metrics are used throughout:

1. the **cosmological metric** $g = \mathrm{diag}(1, -a^2, -a^2, -a^2, -b^2, -b^2, -b^2, -b^2)$ with arbitrary $a(x_0)$ (`a3[x0]`) and $b(x_0)$ (`b4[x0]`);
2. an **inhomogeneous metric** $g = \mathrm{diag}(n^2, -a^2, -a^2, -a^2, -b^2, -b^2, -b^2, -b^2)$ in which n, a, b are explicit polynomials of BOTH x_0 and x_4: $n = 1 + x_0x_4/5 + x_0^2/11$, $a = 1 + x_0/3 + x_4^2/7$, $b = 2 - x_0x_4/4 + x_0^2/9 + x_4/13$ (`nP`, `aP`, `bP`); its identities are checked exactly (rational numbers) at the point $x_0 = 1/2$, $x_4 = 1/3$;
3. a **plane gravitational wave** (pp-wave) $ds^2 = dx_0^2 - dx_1^2 - \cdots - dx_7^2 + F(x_0 - x_1, x_2, x_3)\,(dx_0 - dx_1)^2$ with an arbitrary profile F (`fPP`), which is not diagonal.

For the cosmological metric, with the Hubble rates $H_a = a'/a$ and $H_b = b'/b$, the Riemann tensor in the orthonormal frame is diagonal on index pairs; its only components are the sectional curvatures $K_{AB} = R^{AB}{}_{AB}$ (no sum), with $i, j \in \{1,2,3\}$, $k, l \in \{4,\ldots,7\}$:

$$K_{0i} = -(H_a' + H_a^2), \quad K_{0k} = -(H_b' + H_b^2), \quad K_{ij} = -H_a^2, \quad K_{kl} = -H_b^2, \quad K_{ik} = -H_aH_b.$$

The Einstein tensor and the scalar curvature are

$$\begin{aligned} G^{0}{}_{0} &= 3H_a^2 + 12H_aH_b + 6H_b^2, \\ G^{1}{}_{1} = G^{2}{}_{2} = G^{3}{}_{3} &= 2H_a' + 3H_a^2 + 4H_b' + 10H_b^2 + 8H_aH_b, \\ G^{4}{}_{4} = \cdots = G^{7}{}_{7} &= 3H_a' + 6H_a^2 + 3H_b' + 6H_b^2 + 9H_aH_b, \\ R &= -2\big(3H_a' + 6H_a^2 + 4H_b' + 10H_b^2 + 12H_aH_b\big). \end{aligned}$$

For the inhomogeneous metric the notebook checks, exactly at $x_0 = 1/2$, $x_4 = 1/3$, the symmetries of the Riemann tensor ($R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu} = -R_{\rho\sigma\nu\mu} = R_{\mu\nu\rho\sigma}$ with $R_{\rho\sigma\mu\nu} = g_{\rho\lambda}R^{\lambda}{}_{\sigma\mu\nu}$), the first Bianchi identity $R^{\rho}{}_{\sigma\mu\nu} + R^{\rho}{}_{\mu\nu\sigma} + R^{\rho}{}_{\nu\sigma\mu} = 0$, and the contracted Bianchi identity

$$\nabla_\mu G^{\mu}{}_{\nu} = \partial_\mu G^{\mu}{}_{\nu} + \{^{\mu}_{\mu\lambda}\}G^{\lambda}{}_{\nu} - \{^{\lambda}_{\mu\nu}\}G^{\mu}{}_{\lambda} = 0.$$

(the contracted Bianchi identity is also checked for the cosmological metric with arbitrary $a(x_0)$, $b(x_0)$). A fourth metric enters in section 5.4: the static metric $ds^2 = f(r)\,dx_0^2 - h(r)\,dr^2 - r^2(dx_2^2 + \cdots + dx_7^2)$ with $r = x_1$ and arbitrary positive f, h.

4. The Lovelock tensors of 7+1 spacetime (section 12)

4.1 Lovelock's theorem and the definitions

In n dimensions the most general tensor A^{lh} that is symmetric, divergence-free and built from the metric and its first and second derivatives is (D. Lovelock and H. Rund, *Tensors, Differential Forms, and Variational Principles*, Dover; Remark 2 after their Eq. (4.38))

$$A^{lh} = \sqrt{|g|}\sum_{k=1}^{m-1} \alpha_{(k)}\, g^{jl}\, \delta^{hh_1\cdots h_{2k}}_{jj_1\cdots j_{2k}}\, R^{j_1 j_2}{}_{h_1 h_2}\cdots R^{j_{2k-1}j_{2k}}{}_{h_{2k-1}h_{2k}} + \lambda\sqrt{|g|}\, g^{lh}, \qquad m = \begin{cases} n/2, & n \text{ even},\\ (n+1)/2, & n \text{ odd},\end{cases}$$

with arbitrary constants $\alpha_{(k)}$ and λ. For $n = 8$, $m - 1 = 8/2 - 1 = 3$: there are exactly three curvature terms. The generalized Kronecker delta is the determinant of the matrix of ordinary Kronecker deltas,

$$\delta^{\mu_1\cdots\mu_n}_{\nu_1\cdots\nu_n} = \det\left[\delta^{\mu_i}_{\nu_j}\right]_{i,j=1}^{n}, \qquad \delta^{\mu_1\cdots\mu_p\rho}_{\nu_1\cdots\nu_p\rho} = (8-p)\,\delta^{\mu_1\cdots\mu_p}_{\nu_1\cdots\nu_p} \quad (D=8),$$

it is $+1$ (-1) when the lower indices are an even (odd) permutation of the upper ones, all different, and 0 otherwise; it vanishes when $n > 8$. With the normalization of the reference notebook Einstein-Lovelock-7+1-Nash.nb (P. L. Nash, GPL), which this document follows, the Lovelock tensors (`Lovelock1`, `Lovelock2`, `Lovelock3`) and densities are

$$\text{Lovelock}_k \equiv E^{(k)\mu}{}_{\nu} = -\frac{1}{2^{k+1}}\,\delta^{\mu\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\nu\gamma_1\delta_1\cdots\gamma_k\delta_k}\, R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k},$$

$$L_k = \frac{1}{2^k}\,\delta^{\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\gamma_1\delta_1\cdots\gamma_k\delta_k}\, R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k}.$$

Special cases: $L_0 = 1$, $E^{(0)\mu}{}_{\nu} = -\frac{1}{2}\delta^{\mu}_{\nu}$; $L_1 = R$, $E^{(1)} = G$; $L_2 = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ (Gauss–Bonnet) with $E^{(2)}$ the Lanczos tensor

$$E^{(2)}_{\mu\nu} = 2\big(R\,R_{\mu\nu} - 2R_{\mu\alpha}R^{\alpha}{}_{\nu} - 2R^{\alpha\beta}R_{\mu\alpha\nu\beta} + R_{\mu}{}^{\alpha\beta\gamma}R_{\nu\alpha\beta\gamma}\big) - \tfrac{1}{2}g_{\mu\nu}L_2;$$

L_3, $E^{(3)}$ are cubic in the curvature (section 4.3 gives $E^{(3)}$ explicitly). Because $E^{(k)}$ contains $2k+1$ different upper indices, $E^{(4)} \equiv 0$ in eight dimensions, although L_4 (the Euler density) is not zero; $\sqrt{|g|}L_4$ is a total derivative. Every $E^{(k)}$ is symmetric with both indices down, divergence-free for every metric, and has the trace

$$E^{(k)\mu}{}_{\mu} = -\frac{8-2k}{2}\,L_k.$$

4.2 Action and field equations

The reference notebook writes the Einstein–Lovelock VACUUM field equations as

$$0 = -\Lambda\,\delta^{\mu}_{\nu} + H^{-2}\,w_1\,\text{Lovelock}_1 + H^{-4}\,w_2\,\text{Lovelock}_2 + H^{-6}\,w_3\,\text{Lovelock}_3,$$

with pure numbers w_1, w_2, w_3, Λ and a fundamental inverse length H (it exists because the Lovelock tensors of different orders have different dimensions, length^{-2k}). With matter, and with $\delta(\sqrt{|g|}L_k) = \sqrt{|g|}\,E^{(k)}_{\mu\nu}\,\delta g^{\mu\nu}$ (up to a total derivative) and $\delta S_{\text{matter}} = \frac{1}{2}\int\sqrt{|g|}\,T_{\mu\nu}\,\delta g^{\mu\nu}\,d^8X$, the action

$$S = -\frac{1}{2\kappa}\int d^8X\,\sqrt{|g|}\sum_{k=0}^{3}\alpha_k L_k + S_{\text{matter}}, \qquad \alpha_0 = 2\bar{\Lambda},\quad \alpha_1 = 1,$$

gives the Einstein–Lovelock field equations

$$\sum_{k=0}^{3}\alpha_k\,E^{(k)\mu}{}_{\nu} = \kappa\,T^{\mu}{}_{\nu} \qquad\Longleftrightarrow\qquad G^{\mu}{}_{\nu} + \alpha_2 E^{(2)\mu}{}_{\nu} + \alpha_3 E^{(3)\mu}{}_{\nu} - \bar{\Lambda}\,\delta^{\mu}_{\nu} = \kappa\,T^{\mu}{}_{\nu},$$

with $\kappa = 8\pi G_8$ (G_8 the Newton constant of eight dimensions), the Gauss–Bonnet and cubic couplings α_2, α_3 (dimensions length^2 and length^4) and the cosmological constant $\bar{\Lambda}$ (dimension length^{-2}; it is written Λ in sections 7 and following, where no pure number Λ appears). α_4 does not appear: its tensor vanishes. For $T = 0$ the two forms are the same equation (divide the first by w_1/H^2):

$$\alpha_2 = \frac{w_2}{w_1 H^2}, \qquad \alpha_3 = \frac{w_3}{w_1 H^4}, \qquad \bar{\Lambda} = \frac{\Lambda H^2}{w_1}.$$

4.3 The Lovelock tensors with xAct, as in the reference notebook

Section 12 of 7+1.nb computes the Lovelock tensors exactly as the reference notebook does, with the tensor packages xAct (xPerm, xTensor, xCoba, xTras; J. M. Martín-García et al., GPL; versions xPerm 1.2.3, xTensor 1.2.0, xCoba 0.8.6, xTras 1.4.2). The reference notebook also contains a "Model metric" which its author has marked FAILED, DO NOT USE (it never worked for the 7+1 signature); it is not used anywhere in this project. The set-up (abstract indices a, b, c, d, f, i, j, k, l, m, n, p, q, r, s, y; a metric `g71` with Levi-Civita derivative `CD`; a chart `chartM8` with the coordinate scalars `x0[]`, …, `x7[]`):

```
Needs["xAct`xCoba`"]; Needs["xAct`xTras`"];
$DefInfoQ = False; $CVVerbose = False;
SetOptions[ToCanonical, UseMetricOnVBundle -> None, Verbose -> False];
DefManifold[M8, 8, {a, b, c, d, f, i, j, k, l, m, n, p, q, r, s, y}];
DefMetric[{1, 7, 0}, g71[-a, -b], CD, {";", "D"}, WeightedWithBasis -> AIndex];
DefChart[chartM8, M8, {0, 1, 2, 3, 4, 5, 6, 7}, {x0[], x1[], x2[], x3[], x4[], x5[], x6[],
    ...x7[]},
  ExtendedCoordinateDerivatives -> True, FormatBasis -> {"Partials", "Differentials"}];
MyArrayComponents[expr_] := expr // ToBasis[chartM8] // ComponentArray // ToValues // ToValues
    ...// Simplify;
kδ[lower_, upper_] /; Length[lower] == Length[upper] := Det[Outer[delta, lower, upper]];
```

In the xAct notation the signature is $\{p, q, z\}$ = (number of $+1$, of -1, of 0 eigenvalues). The reference notebook declares $\{7, 1, 0\}$ (the mostly-plus convention); this notebook declares $\{1, 7, 0\}$, which matches its metric $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$. The declaration does not enter the Lovelock tensors with mixed indices, which follow from the components alone, and both declarations give xAct the same sign of the determinant of the metric, $(-1)^q = -1$ for $q = 7$ and for $q = 1$ (`SignDetOfMetric`). While the packages load, their banners and the warnings that xTensor cannot attach rules to `Commutator` and `Anticommutator` (protected built-in symbols of Wolfram 15, not used here) are suppressed. `kδ[lower, upper]` is the generalized Kronecker delta $\delta^{\text{upper}}_{\text{lower}}$ with abstract indices; the Lovelock tensors are then typed exactly as in the reference notebook,

```
Lovelock1 = -(1/4) RiemannCD[-c,-d,f,i] kδ[{-a,-f,-i},{b,c,d}] // Simplification
Lovelock2 = -(1/8) RiemannCD[-c,-d,f,i] RiemannCD[-j,-k,l,m] kδ[{-a,-f,-i,-l,-m},{b,c,d,j,k}]
    ...// Simplification
Lovelock3 = -(1/16) RiemannCD[-c,-d,f,i] RiemannCD[-j,-k,l,m] RiemannCD[-n,-p,q,r]
              kδ[{-a,-f,-i,-l,-m,-q,-r},{b,c,d,j,k,n,p}] // Simplification
```

i.e. $\text{Lovelock}_k = E^{(k)b}{}_a$ with the free indices a (down) and b (up) and $R_{cd}{}^{fi}$ = `RiemannCD[-c,-d,f,i]`. Simplification (xTras) reduces them to curvature monomials; with $R_{ab}{}^{cd}$ the Riemann tensor, $R_a{}^b$ the Ricci tensor and R the scalar curvature the notebook obtains

$$\text{Lovelock}_1 = -\frac{\delta_a{}^b R}{2} + R_a{}^b,$$

$$\begin{aligned}\text{Lovelock}_2 = &-\frac{\delta_a{}^b R^2}{2} + 2\delta_a{}^b R^j{}_k R_j{}^k \\ &- \frac{\delta_a{}^b R_{jk}{}^{lm} R^{jk}{}_{lm}}{2} + 2R R_a{}^b \\ &- 4R_a{}^j R^b{}_j - 4R^j{}_k R_{aj}{}^{bk} \\ &+ 2R_{aj}{}^{kl} R^{bj}{}_{kl},\end{aligned}$$

$$\begin{aligned}\text{Lovelock}_3 = &-\frac{\delta_a{}^b R^3}{2} + 6\delta_a{}^b R R^j{}_k R_j{}^k \\ &- \frac{3\delta_a{}^b R R_{jk}{}^{lm} R^{jk}{}_{lm}}{2} \\ &- 8\delta_a{}^b R_j{}^l R^j{}_k R^k{}_l \\ &- 12\delta_a{}^b R^j{}_k R^l{}_m R_{jl}{}^{km}\end{aligned}$$

$$
\begin{aligned}
&+ 12\delta_a{}^b R^j{}_k R_{jl}{}^{mn} R^{kl}{}_{mn} \\
&+ 4\delta_a{}^b R^{jk}{}_{lm} R_{jn}{}^{lp} R_{kp}{}^{mn} \\
&- \delta_a{}^b R^{jk}{}_{lm} R_{jk}{}^{np} R^{lm}{}_{np} + 3R^2 R_a{}^b \\
&- 12R R_a{}^j R^b{}_j - 12R R^j{}_k R_{aj}{}^{bk} \\
&+ 6R R_{aj}{}^{kl} R^{bj}{}_{kl} + 24 R_j{}^l R^j{}_k R_{al}{}^{bk} \\
&+ 24 R_a{}^j R^b{}_k R_j{}^k + 24 R_a{}^j R^k{}_l R^{bl}{}_{jk} \\
&- 12 R_a{}^j R^{bk}{}_{lm} R_{jk}{}^{lm} - 12 R^j{}_k R_a{}^b R_j{}^k \\
&- 12 R^j{}_k R_{aj}{}^{lm} R^{bk}{}_{lm} \\
&- 24 R^j{}_k R_{al}{}^{km} R^{bl}{}_{jm} \\
&+ 24 R^j{}_k R_{al}{}^{bm} R_{jm}{}^{kl} \\
&+ 3 R_a{}^b R_{jk}{}^{lm} R^{jk}{}_{lm} + 24 R^k{}_l R^b{}_j R_{ak}{}^{jl} \\
&- 12 R^b{}_j R_{ak}{}^{lm} R^{jk}{}_{lm} \\
&- 24 R_{aj}{}^{kl} R^{bm}{}_{kn} R^{jn}{}_{lm} \\
&+ 6 R_{aj}{}^{kl} R^{bj}{}_{mn} R_{kl}{}^{mn} \\
&- 12 R_{kl}{}^{mn} R^{jl}{}_{mn} R_{aj}{}^{bk}.
\end{aligned}
$$

Two checks are made with abstract indices (ToCanonical): $\text{Lovelock}_1 - (R_a{}^b - \frac{1}{2}\delta_a{}^b R) = 0$, and the trace $\text{Lovelock}_{2a}{}^a = -2L_2$ with $L_2 = R^2 - 4R_{cd}R^{cd} + R_{cdfi}R^{cdfi}$. The reference notebook sets the option `UseMetricOnVBundle -> None` of ToCanonical (indices are not moved up and down); for the trace identity the metric must identify the dummy pairs $R_{cd}R^{cd}$ and $R^c{}_d R_c{}^d$, so that one call uses `UseMetricOnVBundle -> All`.

4.4 Components with xAct: CD = LC[g]

For a metric given as an 8×8 matrix of components with lower indices the reference notebook makes it a CTensor in the basis of `chartM8`, lets MetricCompute compute and simplify its curvature, and sets `CD = LC[g]` ("xAct magic": the abstract covariant derivative becomes the Levi-Civita connection of the component metric). After `RiemannCD = Riemann[CD]`, `RicciCD = Ricci[CD]`, `RicciScalarCD = RicciScalar[CD]` and `EinsteinCD = Einstein[CD]` every abstract expression evaluates to components. The notebook function `xActLovelock[gm]` does exactly these steps:

```
xActLovelock[gm_] := Module[{res},
   g71 = CTensor[gm /. toXc, {-chartM8, -chartM8}];
   MetricCompute[g71, chartM8, All, Parallelize -> False, Verbose -> False];
   CD = LC[g71];
   RiemannCD = Riemann[CD]; RicciCD = Ricci[CD]; RicciScalarCD = RicciScalar[CD]; EinsteinCD =
    ...Einstein[CD];
   res = Transpose[asMat8[MyArrayComponents[#]]] & /@ {Lovelock1, Lovelock2, Lovelock3,
    ...EinsteinCD[-a, b]};
   res /. fromXc]
```

Here `toXc` replaces x_μ by the coordinate scalar `xμ[]` and `fromXc` undoes it, `asMat8` turns a vanishing result into the 8×8 zero matrix, and the Transpose makes row μ, column ν hold $E^{(k)\mu}{}_\nu$ (the array of MyArrayComponents has the lower index a first; the pp-wave of section 4.7, whose $E^{(1)\mu}{}_\nu$ is not a symmetric matrix, fixes this order). For the cosmological metric `xActLovelock` gives Lovelock_1 = `EinsteinCD` = the Einstein tensor of section 3, and the closed forms of section 4.6.

4.5 An independent computation: the pair-set algorithm

The same tensors are also computed without xAct. A direct evaluation of the generalized delta would run over 8^{4k+2} index combinations; the notebook functions `pairSets`, `orderedSeqs`, `lovelockE`, `lovelockL` (and tools/lovelock71.py in Python) use the antisymmetries instead. Let $\mathcal{U}$ run over the SETS of k disjoint index pairs ($\alpha<\beta$) not containing μ, and $\mathcal{V}$ over the ORDERED sequences of k disjoint pairs ($\gamma<\delta$) that exhaust $(\mathcal{U}\cup\{\mu\})\setminus\{\nu\}$. Each pair contributes a factor 2 (both orders of a pair give the same term), the $k!$ orderings of the pairs of $\mathcal{U}$ give the same sum, and

$$E^{(k)\mu}{}_{\nu}=-2^{k-1}k!\sum_{\mathcal{U}}\sum_{\mathcal{V}}\mathrm{sgn}(\mu\,\mathcal{U}\to\nu\,\mathcal{V})\prod_{i=1}^{k}R^{\mathcal{V}_i}{}_{\mathcal{U}_i},\qquad L_k=2^k k!\sum_{\mathcal{U}}\sum_{\mathcal{V}}\mathrm{sgn}(\mathcal{U}\to\mathcal{V})\prod_{i=1}^{k}R^{\mathcal{V}_i}{}_{\mathcal{U}_i},$$

where sgn is the sign of the permutation that turns the upper index sequence into the lower one (in Mathematica `Signature[up] Signature[dn]`). For $k=3$ and fixed μ,ν there are 105 upper pair-sets and 90 ordered lower sequences. The notebook compares $E^{(1)}$ with the brute-force contraction $-\frac{1}{4}\delta^{\mu\alpha\beta}_{\nu\gamma\delta}R^{\gamma\delta}{}_{\alpha\beta}$, $E^{(2)}$ with the Lanczos tensor, and $E^{(1)},E^{(2)},E^{(3)}$ of the cosmological metric and of the pp-wave with xAct: the two computations agree exactly. The pair-set sum also works on a table of exact rational numbers (section 4.7).

4.6 Closed forms for the cosmological metric

For the cosmological metric every $E^{(k)}$ is diagonal; the entry $E^{(k)\mu}{}_{\mu}$ (no sum) is $-\frac{1}{2}$ times the density L_k restricted to the index pairs that avoid μ. The results (Ha $=H_a$, dHa $=H_a'$, and so on) are:

$$\begin{aligned}
E^{(1)0}{}_{0}&=3\left(H_a^2+4H_aH_b+2H_b^2\right),\\
E^{(1)1}{}_{1}&=2H_a'+4H_b'+3H_a^2+8H_aH_b+10H_b^2,\\
E^{(1)4}{}_{4}&=3H_a'+3H_b'+6H_a^2+9H_aH_b+6H_b^2.
\end{aligned}$$

$$\begin{aligned}
E^{(2)0}{}_{0}&=-12H_b\left(4H_a^3+18H_a^2H_b+12H_aH_b^2+H_b^3\right),\\
E^{(2)1}{}_{1}&=-32H_a'H_aH_b-48H_a'H_b^2-16H_b'H_a^2-96H_b'H_aH_b\\
&\quad-48H_b'H_b^2-32H_a^3H_b-136H_a^2H_b^2-192H_aH_b^3\\
&\quad-60H_b^4,\\
E^{(2)4}{}_{4}&=-12H_a'H_a^2-72H_a'H_aH_b-36H_a'H_b^2-36H_b'H_a^2\\
&\quad-72H_b'H_aH_b-12H_b'H_b^2-12H_a^4-108H_a^3H_b\\
&\quad-180H_a^2H_b^2-108H_aH_b^3-12H_b^4.
\end{aligned}$$

$$\begin{aligned}
E^{(3)0}{}_{0}&=360H_a^2H_b^3\left(4H_a+3H_b\right),\\
E^{(3)1}{}_{1}&=72H_b^2\left(8H_a'H_aH_b+2H_a'H_b^2+12H_b'H_a^2+8H_b'H_aH_b+8H_a^3H_b+19H_a^2H_b^2+8H_aH_b^3\right),\\
E^{(3)4}{}_{4}&=72H_aH_b\left(9H_a'H_aH_b+6H_a'H_b^2+6H_b'H_a^2+9H_b'H_aH_b+9H_a^3H_b+17H_a^2H_b^2+9H_aH_b^3\right).
\end{aligned}$$

In particular

$$E^{(2)0}{}_{0}=-12H_b\big(4H_a^3+18H_a^2H_b+12H_aH_b^2+H_b^3\big),\qquad E^{(3)0}{}_{0}=360\,H_a^2H_b^3\big(4H_a+3H_b\big).$$

Divergence-freeness becomes, for each k,

$$\frac{d}{dx_0}E^{(k)0}{}_{0}+3H_a\big(E^{(k)0}{}_{0}-E^{(k)1}{}_{1}\big)+4H_b\big(E^{(k)0}{}_{0}-E^{(k)4}{}_{4}\big)=0,$$

checked for arbitrary a,b. The VARIATIONAL ORIGIN is checked as well: with a lapse function $N(x_0)$ ($g_{00}=N^2$) the reduced Lagrangian is $Na^3b^4L_k$, and its Euler–Lagrange expressions at $N=1$ are

$$\frac{\delta(Na^3b^4L_k)}{\delta N}=-2a^3b^4E^{(k)0}{}_{0},\qquad\frac{\delta(Na^3b^4L_k)}{\delta a}=-6a^2b^4E^{(k)1}{}_{1},\qquad\frac{\delta(Na^3b^4L_k)}{\delta b}=-8a^3b^3E^{(k)4}{}_{4},$$

for $k=1,2,3,4$ (for $k=4$ all three vanish identically: $\sqrt{|g|}L_4$ is a total derivative). The factors come from $\delta g^{00}=-2\delta N$, three $\delta g^{ii}=2\delta a/a^3$ and four $\delta g^{kk}=2\delta b/b^3$.

4.7 An inhomogeneous metric and a gravitational wave

For the explicit inhomogeneous metric $g = \mathrm{diag}(n^2, -a^2, -a^2, -a^2, -b^2, -b^2, -b^2, -b^2)$ with $n = 1 + x_0 x_4/5 + x_0^2/11$, $a = 1 + x_0/3 + x_4^2/7$, $b = 2 - x_0 x_4/4 + x_0^2/9 + x_4/13$, the divergence $\nabla_\mu E^{(k)\mu}{}_\nu$ is evaluated EXACTLY (rational arithmetic) at $x_0 = 1/2$, $x_4 = 1/3$ for $k = 1, 2, 3$; it vanishes. The derivative of $E^{(k)}$ (a polynomial of degree k in the curvature) is obtained from the curvature and its first derivatives at the point: $\partial_\lambda E^{(k)} = \frac{d}{d\epsilon} E^{(k)}[R + \epsilon\, \partial_\lambda R]\big|_{\epsilon=0}$. At that point $E^{(2)}$ equals the Lanczos tensor exactly, and

$$E^{(3)0}{}_4\big|_{x_0=1/2,\, x_4=1/3} = q \approx 0.0143928243619,$$

$q \in \mathbb{Q}$ (an exact fraction: numerator of 50, denominator of 52 digits).

For the pp-wave $ds^2 = dx_0^2 - dx_1^2 - \cdots - dx_7^2 + F(x_0 - x_1, x_2, x_3)\,(dx_0 - dx_1)^2$ with an arbitrary profile F and the null vector $l_\mu = (1, -1, 0, \ldots, 0)$,

$$E^{(1)}_{\mu\nu} = \tfrac{1}{2}\big(\partial_2^2 + \partial_3^2\big) F\; l_\mu l_\nu, \qquad E^{(2)} = E^{(3)} = 0 \text{ for every profile } F,$$

so a plane wave whose profile is harmonic in the transverse directions solves the Einstein–Lovelock equations for every α_2, α_3 (with $\bar{\Lambda} = 0$). xAct and the pair-set sum agree for this metric, which is not diagonal.

4.8 Exponential vacuum solutions: eternal inflation of ordinary space and deflation of the auxiliary space

The procedure of the reference notebook — solve the vacuum equations of section 4.2 for w_2, w_3 (and Λ) — is applied to the cosmological metric with constant Hubble rates, $a = e^{H_a x_0}$, $b = e^{H_b x_0}$, $H_a = H$, $H_b = q_H H$ (q_H a pure number; code `qH`). With $H_a' = H_b' = 0$ in the closed forms of section 4.6 every $E^{(k)}$ is a constant diagonal matrix proportional to H^{2k}, and the 64 equations reduce to three distinct ones (components 00, 11, 44). They are not independent: for constant Hubble rates the divergence identity of section 4.6 becomes

$$3H_a\big(\mathrm{eq}_{00} - \mathrm{eq}_{11}\big) + 4H_b\big(\mathrm{eq}_{00} - \mathrm{eq}_{44}\big) = 0,$$

so for every (H_a, H_b) there is a one-parameter family of couplings: given w_1 and Λ the equations fix w_2, w_3. For $q_H = -1$ — ordinary space expands as $e^{H x_0}$ while the auxiliary space contracts as $e^{-H x_0}$, for every time x_0 (eternal inflation of ordinary space with eternal deflation of the auxiliary space) — the three equations are

$$\mathrm{eq}_{00} = -\Lambda - 3w_1 - 36w_2 - 360w_3, \qquad \mathrm{eq}_{11} = -\Lambda + 5w_1 + 28w_2 + 216w_3, \qquad \mathrm{eq}_{44} = -\Lambda + 3w_1 + 12w_2 + 72w_3,$$

and the family is

$$w_2 = \frac{\Lambda}{4} - \frac{w_1}{2}, \qquad w_3 = -\frac{\Lambda}{36} + \frac{w_1}{24};$$

$$\Lambda = 0: \quad \frac{w_2}{w_1} = -\frac{1}{2}, \quad \frac{w_3}{w_1} = \frac{1}{24}, \quad \text{i.e.} \quad \alpha_2 = \frac{w_2}{w_1 H^2} = -\frac{1}{2}\, H^{-2}, \quad \alpha_3 = \frac{w_3}{w_1 H^4} = \frac{1}{24}\, H^{-4}.$$

For general q_H (computed here from the closed forms of section 4.6; the notebook and the Python program obtain the same expressions, and check that they solve all 64 components for every Λ):

$$w_2 = \frac{\big(12q_H^3 + 27q_H^2 + 22q_H + 9\big)\, w_1 - \big(2q_H + 3\big)\, \Lambda}{4q_H\,(6q_H^4 + 36q_H^3 + 66q_H^2 + 51q_H + 16)},$$

$$w_3 = \frac{\big(15q_H^4 + 36q_H^3 + 33q_H^2 + 18q_H + 3\big)\, w_1 - \big(3q_H^2 + 6q_H + 1\big)\, \Lambda}{72q_H^3\,(6q_H^4 + 36q_H^3 + 66q_H^2 + 51q_H + 16)}.$$

5. Generalized frame matrices, canonical spin connection, total covariant derivative (section 13)

5.1 Frame field and generalized frame matrices

A frame field (vielbein) is a square root of the metric, and its inverse is written $e_A{}^\mu$:

$$g_{\mu\nu} = \eta_{AB}\, e^A{}_\mu e^B{}_\nu, \qquad e_A{}^\mu e^A{}_\nu = \delta^\mu_\nu.$$

The GENERALIZED FRAME MATRICES are the position-dependent real 16×16 matrices

$$\Gamma_\mu(X) = e^A{}_\mu(X)\,\Gamma_A, \qquad \Gamma^\mu(X) = e_A{}^\mu(X)\,\Gamma^A, \qquad \Gamma_\mu\Gamma_\nu + \Gamma_\nu\Gamma_\mu = 2g_{\mu\nu}(X)\,1_{16},$$

with $\Gamma^A = \eta^{AA}\Gamma_A$ (no sum). They are real because the frame field and the Γ_A are real.

5.2 The canonical spin connection and the total covariant derivative

The canonical (Levi-Civita) spin connection and its action on 16-component spinors are

$$\omega_\mu{}^A{}_B = e^A{}_\nu\big(\partial_\mu e_B{}^\nu + \{{}^\nu_{\mu\lambda}\} e_B{}^\lambda\big), \qquad \omega_{\mu AB} = \eta_{AC}\,\omega_\mu{}^C{}_B = -\omega_{\mu BA},$$
$$\Omega_\mu = \tfrac14\,\omega_{\mu AB}\,\Gamma^A\Gamma^B, \qquad D_\mu\psi = \partial_\mu\psi + \Omega_\mu\psi, \qquad D_\mu\bar\psi = \partial_\mu\bar\psi - \bar\psi\,\Omega_\mu.$$

The TOTAL covariant derivative acts on every index of an object: with the Christoffel symbols on coordinate indices, with ω on frame indices, and through the commutator with Ω on the two spinor indices of a 16×16 matrix. The canonical spin connection is the one for which the total covariant derivative of the frame field and of every frame matrix vanishes:

$$D_\mu e^A{}_\nu = \partial_\mu e^A{}_\nu - \{{}^\lambda_{\mu\nu}\} e^A{}_\lambda + \omega_\mu{}^A{}_B\, e^B{}_\nu = 0 \qquad \text{(vielbein postulate)},$$
$$\nabla_\mu\Gamma_\nu = \partial_\mu\Gamma_\nu - \{{}^\lambda_{\mu\nu}\}\Gamma_\lambda + [\Omega_\mu, \Gamma_\nu] = 0,$$
$$D_\mu\Gamma^A = \omega_\mu{}^A{}_B\,\Gamma^B + [\Omega_\mu, \Gamma^A] = 0.$$

Ω_μ is real and $\Gamma_0\Omega_\mu^{\mathsf T}\Gamma_0 = -\Omega_\mu$, which makes $D_\mu\bar\psi$ the Dirac adjoint of $D_\mu\psi$ and the action below real.

5.3 Spin curvature, Bianchi identity, Lichnerowicz

The curvature of the spin connection is the Riemann tensor in spinor form:

$$\mathcal{F}_{\mu\nu} = \partial_\mu\Omega_\nu - \partial_\nu\Omega_\mu + [\Omega_\mu, \Omega_\nu] = [D_\mu, D_\nu] = \tfrac14\, R_{AB\mu\nu}\,\Gamma^A\Gamma^B, \qquad R_{AB\mu\nu} = \eta_{AC}\, e^C{}_\rho\, R^\rho{}_{\sigma\mu\nu}\, e_B{}^\sigma,$$
$$0 = \partial_\lambda\mathcal{F}_{\mu\nu} + [\Omega_\lambda, \mathcal{F}_{\mu\nu}] + \partial_\mu\mathcal{F}_{\nu\lambda} + [\Omega_\mu, \mathcal{F}_{\nu\lambda}] + \partial_\nu\mathcal{F}_{\lambda\mu} + [\Omega_\nu, \mathcal{F}_{\lambda\mu}] \qquad \text{(Bianchi)},$$
$$\tfrac12\,\Gamma^\mu\Gamma^\nu\mathcal{F}_{\mu\nu} = -\tfrac{R}{4}\,1_{16}, \qquad (\Gamma^\mu D_\mu)^2\psi = g^{\mu\nu}\big(\partial_\mu D_\nu\psi + \Omega_\mu D_\nu\psi - \{{}^\lambda_{\mu\nu}\} D_\lambda\psi\big) - \tfrac{R}{4}\psi.$$

Every identity of sections 5.1–5.3 is checked by the notebook (function `frameChecks`) and by the Python program for four metrics: the cosmological metric with the diagonal frame $e = \mathrm{diag}(1, a, a, a, b, b, b, b)$; the pp-wave with the non-diagonal frame $e^0 = (1+F/2)dx_0 - (F/2)dx_1$, $e^1 = (F/2)dx_0 + (1-F/2)dx_1$, $e^A = dx^A$ otherwise; the inhomogeneous metric with $e = \mathrm{diag}(n, a, a, a, b, b, b, b)$ (exactly at $x_0 = 1/2$, $x_4 = 1/3$, the derivatives being taken symbolically first); and the static radial metric of section 5.4.

5.4 Cartan's first structure equation; a static metric with a radial coordinate

With the frame one-forms $e^A = e^A{}_\mu dx^\mu$ and the connection one-forms $\omega^A{}_B = \omega_\mu{}^A{}_B\, dx^\mu$, the vielbein postulate antisymmetrized in μ, ν (the Christoffel terms cancel) is Cartan's first structure equation without torsion:

$$de^A + \omega^A{}_B \wedge e^B = 0 \quad\Longleftrightarrow\quad \partial_\mu e^A{}_\nu - \partial_\nu e^A{}_\mu + \omega_\mu{}^A{}_B\, e^B{}_\nu - \omega_\nu{}^A{}_B\, e^B{}_\mu = 0.$$

It determines the canonical spin connection from the frame alone. With the structure coefficients $C^A{}_{BC} = -C^A{}_{CB}$ and the connection written with a frame form index,

$$de^A = \tfrac12\, C^A{}_{BC}\, e^B \wedge e^C, \qquad C^A{}_{BC} = e_B{}^\mu e_C{}^\nu\big(\partial_\mu e^A{}_\nu - \partial_\nu e^A{}_\mu\big), \qquad C_{ABC} = \eta_{AD} C^D{}_{BC},$$

$$\omega_{C,AB} \equiv e_C{}^\mu\,\omega_{\mu AB} = -\tfrac{1}{2}\big(C_{ACB} + C_{BAC} - C_{CBA}\big),$$

the unique solution with $\omega_{C,AB} = -\omega_{C,BA}$; both relations are checked for all four frames. As an example take $r = x_1 > 0$ and arbitrary positive functions $f(r)$, $h(r)$ (code `fR[x1]`, `hR[x1]`):

$$ds^2 = f(r)\,dx_0^2 - h(r)\,dr^2 - r^2\big(dx_2^2 + dx_3^2 + dx_4^2 + dx_5^2 + dx_6^2 + dx_7^2\big),$$
$$e^0 = \sqrt{f}\,dx_0, \quad e^1 = \sqrt{h}\,dr, \quad e^k = r\,dx^k \ (k = 2, \ldots, 7).$$

The six directions $x_2, \ldots, x_7$ are flat directions scaled by r (a static "planar" metric of 7+1 spacetime, not a spherically symmetric one). Cartan's equation gives $de^0 = \frac{f'}{2\sqrt{f}}\,dr \wedge dx_0 = -\omega^0{}_1 \wedge e^1$ and $de^k = dr \wedge dx^k = -\omega^k{}_1 \wedge e^1$, hence $\omega^0{}_1 = \frac{f'}{2\sqrt{fh}}\,dx_0$, $\omega^k{}_1 = \frac{1}{\sqrt{h}}\,dx^k$, and

$$\Omega_0 = \frac{f'}{4\sqrt{f}\sqrt{h}}\,\Gamma_1\Gamma_0, \qquad \Omega_1 = 0, \qquad \Omega_k = \frac{1}{2\sqrt{h}}\,\Gamma_1\Gamma_k \quad (k = 2, \ldots, 7),$$
$$\Gamma^\mu\Omega_\mu = \frac{1}{\sqrt{h}}\Big(\frac{f'}{4f} + \frac{3}{r}\Big)\Gamma^1, \qquad \Gamma^1 = \eta^{11}\Gamma_1 = -\Gamma_1,$$

the $3/r = 6 \times \frac{1}{2r}$ coming from the six scaled directions. The covariant Dirac operator and a rescaling that removes the connection are therefore

$$i\Gamma^\mu D_\mu - m = \frac{i}{\sqrt{f}}\,\Gamma^0\partial_0 + \frac{i}{\sqrt{h}}\,\Gamma^1\Big(\partial_r + \frac{f'}{4f} + \frac{3}{r}\Big) + \sum_{k=2}^{7}\frac{i}{r}\,\Gamma^k\partial_k - m,$$
$$\psi = f^{-1/4}r^{-3}\chi: \qquad \Gamma^\mu D_\mu\psi = f^{-1/4}r^{-3}\,\Gamma^\mu\partial_\mu\chi.$$

6. The covariant Dirac equation of dirac16complex

The Lagrangian density and the field equation are

$$\mathcal{L} = \sqrt{|g|}\;\bar\psi\big(i\,\Gamma^\mu D_\mu - m\big)\psi, \qquad \big(i\,\Gamma^\mu(X)\,D_\mu - m\big)\psi = 0,$$

where the spin connection Ω_μ in D_μ is the coupling of gravity to dirac16complex. The current $J^\mu = \bar\psi\Gamma^\mu\psi$ is conserved because, for arbitrary ψ and $\bar\psi$ (by $\nabla\Gamma = 0$ and $\partial_\mu\sqrt{|g|} = \sqrt{|g|}\{{}^{\lambda}_{\lambda\mu}\}$),

$$\partial_\mu\big(\sqrt{|g|}\,\bar\psi\Gamma^\mu\psi\big) = \sqrt{|g|}\,\big[(D_\mu\bar\psi)\Gamma^\mu\psi + \bar\psi\Gamma^\mu D_\mu\psi\big],$$

and the bracket vanishes when ψ and $\bar\psi$ solve the field equation and its adjoint. Squaring with the Lichnerowicz formula gives $\big(g^{\mu\nu}\nabla_\mu D_\nu + m^2 - \frac{R}{4}\big)\psi = 0$.

Coupling to a gauge field. An abelian gauge field $A_\mu(X)$ with charge q enters through the gauge-covariant derivative

$$D_\mu\psi = \partial_\mu\psi + \Omega_\mu\psi - iqA_\mu\psi, \qquad D_\mu\bar\psi = \partial_\mu\bar\psi - \bar\psi\,\Omega_\mu + iqA_\mu\bar\psi.$$

Under $\psi \to e^{iq\alpha}\psi$, $A_\mu \to A_\mu + \partial_\mu\alpha$ ($\alpha(X)$ arbitrary) the Dirac operator is covariant, $\Gamma^\mu D'_\mu(e^{iq\alpha}\psi) = e^{iq\alpha}\,\Gamma^\mu D_\mu\psi$, and the current identity above holds unchanged, so the electric current $q\,\bar\psi\Gamma^\mu\psi$ is conserved. Both statements are checked with the cosmological frame and arbitrary $A_\mu(x_0, x_1, x_5)$, $\alpha(x_0, x_1, x_5)$.

6.1 The cosmological spacetime

For the cosmological metric the spin connection is NOT zero whenever space expands:

$$\Omega_0 = 0, \qquad \Omega_i = \tfrac{a'}{2}\,\Gamma_i\Gamma_0 \ (i = 1, 2, 3), \qquad \Omega_k = \tfrac{b'}{2}\,\Gamma_k\Gamma_0 \ (k = 4, \ldots, 7), \qquad \Gamma^\mu\Omega_\mu = \Big(\tfrac{3}{2}H_a + 2H_b\Big)\Gamma_0.$$

The rescaled field $\chi = (a^3b^4)^{1/2}\psi$ obeys an equation without connection, $\Gamma^\mu D_\mu\psi = (a^3b^4)^{-1/2}\Gamma^\mu\partial_\mu\chi$. For a plane wave $\chi = e^{i(\vec p\cdot\vec x + \vec q\cdot\vec y)}c(x_0)$ — $\vec p = (p_1, p_2, p_3)$ the momentum along ordinary space $\vec x = (x_1, x_2, x_3)$, $\vec q = (q_4, \ldots, q_7)$ along the auxiliary space $\vec y = (x_4, \ldots, x_7)$ — the Dirac equation becomes

$$i\,\frac{dc}{dx_0} = h(x_0)\,c, \qquad h(x_0) = \sum_{i=1}^{3}\alpha_i\,\frac{p_i}{a} + \sum_{k=4}^{7}\alpha_k\,\frac{q_k}{b} + \beta\,m, \qquad \beta = \Gamma_0, \quad \alpha_j = \Gamma^0\Gamma^j = \begin{pmatrix} 0 & \tau_j^O \\ -\tau_j^O & 0 \end{pmatrix},$$

the flat-space Hamiltonian with the physical (redshifted) momenta $\vec p/a$ and $\vec q/b$.

6.2 Conformal weight 7/2

For a conformally flat metric $g = C(x_0)^2\eta$ (x_0 is then conformal time) the massless operator satisfies, for every f,

$$\Gamma^\mu D_\mu\big(C^{-7/2} f\big) = C^{-9/2}\,\Gamma^A \partial_A f,$$

so a massless dirac16complex in a conformally flat 7+1 spacetime is a rescaled flat-space solution and no pairs are created. The cosmological metric is conformally flat only when $a = b$.

6.3 Two classes of states in an anisotropic expansion

With unit vectors $\vec{n}_a$, $\vec{n}_b$ along $\vec{p}$ and $\vec{q}$ the Hamiltonian is $h = \frac{|\vec{p}|}{a}\mathcal{A} + \frac{|\vec{q}|}{b}\mathcal{B} + m\beta$ with the CONSTANT real matrices $\mathcal{A} = \sum_i n_{a,i}\alpha_i$, $\mathcal{B} = \sum_k n_{b,k}\alpha_k$. $\mathcal{A}$, $\mathcal{B}$, β square to 1_{16} and anticommute; $\mathcal{Z} = \mathcal{A}\mathcal{B}\beta$ squares to -1_{16}, is antisymmetric, and commutes with $\mathcal{A}$, $\mathcal{B}$, β and h, so its eigenvalue $\pm i$ is conserved. On $\mathcal{Z} = +i$ the three matrices act like the Pauli matrices $(\sigma_1, \sigma_2, \sigma_3)$, on $\mathcal{Z} = -i$ like $(\sigma_1, \sigma_2, -\sigma_3)$; the 16-component mode equation is four copies of each of

$$i\,\frac{dc}{dx_0} = \Big(\frac{|\vec{p}|}{a}\sigma_1 + \frac{|\vec{q}|}{b}\sigma_2 \pm m\,\sigma_3\Big)c, \qquad c \in \mathbb{C}^2.$$

Pairs are created when the direction of the physical momentum $(|\vec{p}|/a, |\vec{q}|/b)$ turns fast compared with the frequency $\omega = \sqrt{|\vec{p}|^2/a^2 + |\vec{q}|^2/b^2 + m^2}$. The Bogoliubov coefficient $|\beta|^2$ of a mode is the probability that a positive-energy state at the initial time ends in a negative-energy state at the final time; $|\alpha|^2 + |\beta|^2 = 1$.

7. The three epochs (section 14)

For the cosmological metric with $T^\mu{}_\nu = \mathrm{diag}(\varepsilon, -p_a, -p_a, -p_a, -p_b, -p_b, -p_b, -p_b)$ (energy density ε, pressures p_a along ordinary space and p_b along the auxiliary space) the field equations are

$$\sum_k \alpha_k E^{(k)0}{}_0 = \kappa\varepsilon, \qquad \sum_k \alpha_k E^{(k)1}{}_1 = -\kappa p_a, \qquad \sum_k \alpha_k E^{(k)4}{}_4 = -\kappa p_b,$$

and they imply the conservation law $\varepsilon' + 3H_a(\varepsilon + p_a) + 4H_b(\varepsilon + p_b) = 0$.

7.1 Primordial epoch (near $x_0 = 0$): Kasner solutions and dynamical compactification

For $a = x_0^p$, $b = x_0^q$ every term of $E^{(k)}$ scales as x_0^{-2k}: near $x_0 = 0$ the highest Lovelock order with a non-zero coupling dominates, and matter whose density grows more slowly than x_0^{-2k} is negligible. With $H_a = p/x_0$, $H_b = q/x_0$, $H_a' = -p/x_0^2$, $H_b' = -q/x_0^2$ in the closed forms of section 4.6 the vacuum solutions of each order are:

- **Einstein** ($k = 1$): $3p + 4q = 1$ and $3p^2 + 4q^2 = 1$, i.e. $p = (1 \pm 2\sqrt{2})/7$, $q = (2 \mp 3\sqrt{2})/14$. On the branch $p = 0.5469$, $q = -0.1602$ ordinary space expands while the auxiliary space contracts as x_0 grows: a dynamical compactification of $x_4, \ldots, x_7$.
- **Gauss–Bonnet** ($k = 2$): $3p+4q = 3$, and p is a root of $-203p^3+81p^2+351p+27 = 0$: $p = -1.0835018$, -0.078631058, 1.5611476 (with $q = (3 - 3p)/4$).
- **Cubic Lovelock** ($k = 3$): the non-degenerate solution $(p, q) = (-15/7,\ 20/7)$ with $3p + 4q = 5$; $E^{(3)}$ vanishes identically on the degenerate families $q = 0$ (static auxiliary space) and $p = 0$.

The pure-order equations are (each multiplied by x_0^{2k}; columns 00, 11, 44):

$$\begin{aligned} x_0^2 E^{(1)0}{}_0 &= 3\left(p^2 + 4pq + 2q^2\right), \\ x_0^2 E^{(1)1}{}_1 &= 3p^2 + 8pq - 2p + 10q^2 - 4q, \\ x_0^2 E^{(1)4}{}_4 &= 3\left(2p^2 + 3pq - p + 2q^2 - q\right). \end{aligned}$$

$$x_0^4 E^{(2)0}{}_0 = -12q\left(4p^3+18p^2q+12pq^2+q^3\right),$$
$$x_0^4 E^{(2)1}{}_1 = -4q\left(8p^3+34p^2q-12p^2+48pq^2-36pq+15q^3-12q^2\right),$$
$$x_0^4 E^{(2)4}{}_4 = -12\left(p^4+9p^3q-p^3+15p^2q^2-9p^2q+9pq^3-9pq^2+q^4-q^3\right).$$

$$x_0^6 E^{(3)0}{}_0 = 360p^2q^3\,(4p+3q),$$
$$x_0^6 E^{(3)1}{}_1 = 72pq^3\left(8p^2+19pq-20p+8q^2-10q\right),$$
$$x_0^6 E^{(3)4}{}_4 = 72p^2q^2\left(9p^2+17pq-15p+9q^2-15q\right).$$

7.2 Pair creation in the primordial epoch

On the Einstein–Kasner branch a massless mode depends on $(|\vec{p}|,|\vec{q}|)$ only through the scaling variable $K=|\vec{p}|\,\lambda^{1-p}$, $\lambda=(|\vec{q}|/|\vec{p}|)^{1/(q-p)}$ ($x_0\to\lambda x_0$ maps one problem onto the other). The evolution is adiabatic before and after the time at which $|\vec{p}|/a\approx|\vec{q}|/b$, so it is integrated from $10^{-15}\lambda$ to 300λ:

$(\lvert\vec{p}\rvert,\lvert\vec{q}\rvert)$	K	$\lvert\beta\rvert^2$ **(Mathematica)**	$\lvert\beta\rvert^2$ **(Python)**
$(1,1)$	1.00000	0.010749346	0.010749346
$(2,1)$	3.11829	0.00013294108	0.00013294108
$(1,3)$	0.494632	0.067442022	0.067442022

The momenta $(1.4302\ldots,2.5)$ have $K=1$ and give the same $|\beta|^2$ as $(1,1)$; an isotropic expansion creates no massless pairs ($|\beta|^2<10^{-20}$). With a mass the two classes $\mathcal{Z}=\pm i$ are created at different rates: for $m=0.3$, $|\vec{p}|=1$, $|\vec{q}|=1.5$, x_0 from 0.05 to 40,

$$|\beta|^2_{16\times16}=0.027052647=\tfrac{1}{2}\big(|\beta|^2_{+m}+|\beta|^2_{-m}\big),\qquad |\beta|^2_{+m}=0.025286593,\quad |\beta|^2_{-m}=0.028818702.$$

7.3 Modern epoch: a static auxiliary space

With $b=b_0$ ($H_b=0$) the Lovelock tensors reduce to (entries 00, 11, 44)

$$E^{(1)}=\mathrm{diag}\big(3H_a^2,\ 2H_a'+3H_a^2,\ 3H_a'+6H_a^2\big),\qquad E^{(2)}=\mathrm{diag}\big(0,\ 0,\ -12H_a^2(H_a'+H_a^2)\big),\qquad E^{(3)}=0.$$

The 00 and 11 equations are the Friedmann equations of ordinary cosmology,

$$3H_a^2=\kappa\varepsilon+\Lambda=8\pi G_4\,\rho_4+\Lambda,\qquad 2H_a'+3H_a^2=-\kappa p_a+\Lambda,\qquad G_4=\frac{G_8}{V_\perp},\quad V_\perp=b_0^4V_c,$$

for matter confined to ordinary space ($\varepsilon=\rho_4/V_\perp$; V_c the coordinate volume of the auxiliary space). The 44 equation then FIXES the stress that holds b constant: $\kappa p_b=\Lambda-3H_a'-6H_a^2+12\alpha_2H_a^2(H_a'+H_a^2)$ — the analogue of the fold fluid of Dirac_claude. A mode with auxiliary momentum $\vec{q}$ moves in ordinary space as a particle of mass $\sqrt{m^2+|\vec{q}|^2/b_0^2}$ (a Kaluza–Klein tower on a torus). Pair creation today is suppressed by $e^{-2\pi M/H_0}$ (section 7.4) with $H_0\approx1.4\times10^{-42}$ GeV, i.e. absent.

7.4 Far future: de Sitter expansion and thermal pair creation

Eight-dimensional de Sitter space $a=b=e^{Hx_0}$ is an exact vacuum solution when all $E^{(k)}$ are proportional to δ:

$$E^{(1)}=21H^2\,\delta,\quad E^{(2)}=-420H^4\,\delta,\quad E^{(3)}=2520H^6\,\delta\qquad\Longrightarrow\qquad 21\alpha_1H^2-420\alpha_2H^4+2520\alpha_3H^6=\Lambda.$$

Over a static auxiliary space ($a=e^{Hx_0}$, $b=b_0$): $E^{(1)}=\mathrm{diag}(3,3,6)H^2$, $E^{(2)}=\mathrm{diag}(0,0,-12)H^4$, $E^{(3)}=0$. In both cases a mode of dirac16complex obeys $i\,dc/dx_0=(k\,e^{-Hx_0}\sigma_1+M\sigma_3)c$, with $M=m$, or with the Kaluza–Klein mass $M=\sqrt{m^2+|\vec{q}|^2/b_0^2}$. Starting from the adiabatic vacuum at early times the number

of created pairs per mode is exactly the Fermi–Dirac occupation at the Gibbons–Hawking temperature $T = H/(2\pi)$, independent of k:

$$|\beta|^2 = \frac{1}{e^{2\pi M/H} + 1}.$$

M/H	$\lvert\beta\rvert^2$ **(Mathematica)**	$1/(e^{2\pi M/H} + 1)$
0.25	0.172102919	0.172102899
0.5	0.0414238845	0.0414238322
1.0	0.00186397746	0.00186396189

The integrator is also checked on the Landau–Zener problem $i\,dc/dt = (\kappa\sigma_1 + t\sigma_3)c$, t from -150 to 150, whose exact result is $|\beta|^2 = e^{-\pi\kappa^2}$.

8. Verification

Every statement above is a check of 7+1.nb (Wolfram Language 15.0.1) and of the independent Python program tools/ref_gravity.py (sympy, numpy, scipy), which rebuilds the Dirac matrices from 4+4.nb's split octonions and computes everything again with its own curvature and Lovelock engine (tools/lovelock71.py). The checks of sections 11–14 of the notebook:

#	check	result	section
1	cosmological metric: sectional curvatures K_0i = -(Ha'+Ha^2), K_0k = -(Hb'+Hb^2), K_ij = -Ha^2, K_kl = -Hb^2, K_ik = -Ha Hb	True	11.2
2	cosmological metric: R^cd_ab is non-zero only for {c,d} = {a,b}	True	11.2
3	Einstein tensor: G^0_0 = 3Ha^2 + 12 Ha Hb + 6 Hb^2, G^1_1 = 2Ha' + 3Ha^2 + 4Hb' + 10Hb^2 + 8HaHb, G^4_4 = 3Ha' + 6Ha^2 + 3Hb' + 6Hb^2 + 9HaHb	True	11.2
4	scalar curvature R = -2 (3Ha' + 6Ha^2 + 4Hb' + 10Hb^2 + 12 Ha Hb)	True	11.2
5	inhomogeneous metric, exactly at (x0, x4) = (1/2, 1/3): R_rsmn = -R_srmn = -R_rsnm = R_mnrs and R^r_[smn] = 0 (first Bianchi identity)	True	11.3
6	inhomogeneous metric: contracted Bianchi identity nabla_mu G^mu_nu == 0, exactly at (x0, x4) = (1/2, 1/3)	True	11.3
7	cosmological metric: contracted Bianchi identity nabla_mu G^mu_nu = 0 (arbitrary a, b)	True	11.3
8	generalized Kronecker delta: contraction identity delta^{m1..mp r}_{n1..np r} = (8-p) delta^{m1..mp}_{n1..np} (p = 1..4, 60 random index sets each)	True	12.1
9	generalized Kronecker delta with 9 upper indices vanishes in 8 dimensions (100 random index sets)	True	12.1
10	xAct: Lovelock1 == R_a^b - (1/2) delta_a^b R, the Einstein tensor (abstract indices)	True	12.3
11	xAct: trace of Lovelock2 == -2 L_2 with L_2 = R^2 - 4 R_cd R^cd + R_cdfi R^cdfi (abstract indices)	True	12.3
12	xAct, cosmological metric: Lovelock1 == EinsteinCD == the Einstein tensor of the curvature engine of section 11.1	True	12.4
13	pair-set algorithm == brute-force generalized-delta contraction for E^(1) (cosmological metric)	True	12.5
14	pair-set algorithm == xAct (section 12.4) for E^(1), E^(2), E^(3) of the cosmological metric (two independent computations)	True	12.5
15	E^(1) == Einstein tensor G and L_1 == R (cosmological metric)	True	12.6
16	trace identity E^(k)mu_mu == -((8-2k)/2) L_k for k = 0..4	True	12.6
17	E^(4) == 0 identically in 8 dimensions, while the Euler density L_4 != 0	True	12.6
18	every E^(k) is diagonal, E^(k)1_1 = E^(k)2_2 = E^(k)3_3 and E^(k)4_4 = ... = E^(k)7_7	True	12.6
19	E^(2) == Lanczos tensor and L_2 == R^2 - 4 Ric.Ric + Riem.Riem (cosmological metric)	True	12.6
20	E^(2)0_0 = -12 Hb (4Ha^3 + 18Ha^2 Hb + 12 Ha Hb^2 + Hb^3) and E^(3)0_0 = 360 Ha^2 Hb^3 (4Ha + 3Hb)	True	12.6
21	divergence-free: d0 E^0_0 + 3Ha (E^0_0 - E^1_1) + 4Hb (E^0_0 - E^4_4) == 0 for k = 1, 2, 3 (arbitrary a, b)	True	12.6
22	variation of nL a^3 b^4 L_k: EL_nL = -2 a^3 b^4 E^(k)0_0, EL_a = -6 a^2 b^4 E^(k)1_1, EL_b = -8 a^3 b^3 E^(k)4_4 (k = 1..4)	True	12.6
23	inhomogeneous metric: nabla_mu E^(k)mu_nu == 0 EXACTLY at (x0, x4) = (1/2, 1/3) for k = 1, 2, 3	True	12.7

#	check	result	section
24	inhomogeneous metric: E^(1) == G at the point, and E^(k)_mu nu symmetric (k = 1, 2, 3)	True	12.7
25	inhomogeneous metric: E^(2) == Lanczos tensor at the point (exact)	True	12.7
26	pp-wave: E^(1)_mu nu == +(1/2)(d2^2 + d3^2) fPP l_mu l_nu	True	12.8
27	pp-wave: E^(2) == E^(3) == 0 for every profile fPP	True	12.8
28	pp-wave: xAct (section 12.4) == pair-set algorithm for E^(1), E^(2), E^(3) (a metric that is not diagonal; arbitrary profile fPP)	True	12.8
29	exponential expansion a = e^{H x0}, b = e^{qH H x0}: xAct Lovelock_k == the closed forms of section 12.6 with Ha = H, Hb = qH H, Ha' = Hb' = 0 (k = 1, 2, 3)	True	12.9
30	exponential expansion: three distinct vacuum equations, related by 3 Ha (eq00 - eq11) + 4 Hb (eq00 - eq44) == 0	True	12.9
31	exponential vacuum solutions: (w2, w3) solve all 64 components for every qH (generic qH) and every Lambda	True	12.9
32	eternal inflation (a = e^{H x0}) with deflation (b = e^{-H x0}): w2 == Lambda/4 - w1/2, w3 == w1/24 - Lambda/36; for Lambda = 0: w2/w1 == -1/2, w3/w1 == 1/24	True	12.9
33	cosmological metric: Omega_0 == 0, Omega_i == (a'/2) Gamma17[i].Gamma17[0], Omega_k == (b'/2) Gamma17[k].Gamma17[0]	True	13.2
34	cosmological metric: Gamma^mu Omega_mu == (3a'/(2a) + 2b'/b) Gamma17[0]	True	13.2
35	current identity: d_mu(sqrt|g| psibar Gamma^mu psi) == sqrt|g| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi] (psi, psibar arbitrary)	True	13.2
36	Lichnerowicz formula: (Gamma^mu D_mu)^2 psi == (g^mu nu nabla_mu D_nu - R/4) psi	True	13.2
37	rescaling psi = (a^3 b^4)^(-1/2) chi removes the spin connection: Gamma^mu D_mu psi == (a^3 b^4)^(-1/2) Gamma^mu d_mu chi	True	13.2
38	mode equation: Gamma17[0].(i Gamma^mu D_mu - m) psi == (a^3 b^4)^(-1/2) e^{i(p.x+q.y)} (i dc/dx0 - h(x0) c) for psi = (a^3 b^4)^(-1/2) e^{i(p.x+q.y)} c(x0)	True	13.2
39	conformal weight 7/2: Gamma^mu D_mu (C^(-7/2) f) == C^(-9/2) Gamma17up[A] d_A f for g = C(x0)^2 eta17	True	13.4
40	Cartan's first structure equation de^A + omega^A_B ^ e^B == 0 (no torsion): cosmological, pp-wave and inhomogeneous frames	True	13.5
41	structure coefficients: omega_C,AB == -(1/2)(C_ACB + C_BAC - C_CBA) with C^A_BC = e_B^mu e_C^nu (d_mu e^A_nu - d_nu e^A_mu): cosmological and inhomogeneous frames	True	13.5
42	static radial metric: Cartan's first structure equation and omega_C,AB == -(1/2)(C_ACB + C_BAC - C_CBA)	True	13.5
43	static radial metric: Omega_0 == (f'/(4 Sqrt[f] Sqrt[h])) Gamma17[1].Gamma17[0], Omega_1 == 0, Omega_k == (1/(2 Sqrt[h])) Gamma17[1].Gamma17[k] (k = 2..7)	True	13.5
44	static radial metric: Gamma^mu Omega_mu == (1/Sqrt[h]) (f'/(4 f) + 3/r) Gamma17up[1]	True	13.5
45	static radial metric: psi = f^(-1/4) r^(-3) chi removes the spin connection, Gamma^mu D_mu psi == f^(-1/4) r^(-3) Gamma^mu d_mu chi	True	13.5
46	U(1) gauge covariance: Gamma^mu (D_mu - i q A'_mu)(e^{i q alpha} psi) == e^{i q alpha} Gamma^mu (D_mu - i q A_mu) psi with A'_mu = A_mu + d_mu alpha	True	13.5
47	current identity with the gauge field: d_mu(sqrt|g| psibar Gamma^mu psi) == sqrt|g| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi], D including -i q A_mu	True	13.5
48	Kasner metric: the Einstein tensor computed directly from the metric equals the closed form (x0^2 G^mu_mu, mu = 0, 1, 4)	True	14.1
49	Einstein Kasner: (p, q) = ((1+2Sqrt[2])/7, (2-3Sqrt[2])/14) and ((1-2Sqrt[2])/7, (2+3Sqrt[2])/14) solve E^(1) == 0	True	14.1
50	Gauss-Bonnet Kasner: 3 real roots of the cubic; (p, (3-3p)/4) solve E^(2) == 0	True	14.1
51	cubic Lovelock Kasner: (p, q) = (-15/7, 20/7) solves E^(3) == 0; E^(3) == 0 identically for q = 0 and for p = 0	True	14.1
52	2x2 reduction: A, B, betaD square to ID16 and anticommute; Zc = A.B.betaD: Zc.Zc == -ID16, Transpose[Zc] == -Zc, Zc commutes with A, B, betaD	True	14.2
53	Kasner pair creation: |alpha|^2 + |beta|^2 == 1 (to 10^-8) for (|p|, |q|) = (1,1), (2,1), (1,3)	True	14.3
54	Kasner scaling: (|p|, |q|) = (1.4302.., 2.5) has K = 1 and the same |beta|^2 as (1, 1) (to 10^-6)	True	14.3
55	16x16 evolution == 4 copies of the 2x2 evolution with +m sigma3 and 4 with -m sigma3 (Kasner, m = 0.3, |p| = 1, |q| = 1.5, x0 from 0.05 to 40): average |beta|^2 agrees to 10^-8; unitarity; the two classes differ	True	14.3
56	isotropic expansion, massless: no pair creation (|beta|^2 < 10^-20)	True	14.3
57	static auxiliary space (Hb = 0): E^(1) = (3Ha^2, 2Ha' + 3Ha^2, 3Ha' + 6Ha^2), E^(2) = (0, 0, -12 Ha^2 (Ha' + Ha^2)), E^(3) = 0	True	14.4
58	8D de Sitter: E^(1) = 21 H^2, E^(2) = -420 H^4, E^(3) = 2520 H^6 (all diagonal entries)	True	14.5
59	de Sitter over a static auxiliary space: E^(1) = (3, 3, 6) H^2, E^(2) = (0, 0, -12) H^4, E^(3) = 0	True	14.5
60	de Sitter pair creation: |beta|^2 == 1/(Exp[2 Pi M/H] + 1) to 10^-7, M/H = 0.25, 0.5, 1	True	14.5
61	integrator benchmark (Landau-Zener): |beta|^2 == Exp[-Pi kappa^2] to 10^-6	True	14.5

#	check	result	section
62	cosmological metric: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True	13.1
63	cosmological metric: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True	13.1
64	cosmological metric: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True	13.1
65	cosmological metric: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True	13.1
66	cosmological metric: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True	13.1
67	cosmological metric: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True	13.1
68	cosmological metric: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True	13.1
69	cosmological metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True	13.1
70	cosmological metric: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True	13.1
71	pp-wave: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True	13.1
72	pp-wave: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True	13.1
73	pp-wave: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True	13.1
74	pp-wave: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True	13.1
75	pp-wave: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True	13.1
76	pp-wave: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True	13.1
77	pp-wave: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True	13.1
78	pp-wave: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True	13.1
79	pp-wave: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True	13.1
80	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True	13.1
81	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True	13.1
82	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True	13.1
83	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True	13.1
84	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True	13.1
85	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True	13.1
86	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True	13.1
87	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True	13.1
88	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True	13.1
89	static radial metric: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True	13.1
90	static radial metric: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True	13.1
91	static radial metric: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True	13.1
92	static radial metric: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True	13.1
93	static radial metric: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True	13.1
94	static radial metric: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True	13.1
95	static radial metric: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True	13.1
96	static radial metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True	13.1
97	static radial metric: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True	13.1

The Python program's boolean checks:

#	check	result
1	cosmological metric: sectional curvatures R^{AB}_{AB} = -(Ha'+Ha^2), -(Hb'+Hb^2), -Ha^2, -Hb^2, -Ha Hb	True
2	cosmological metric: Riemann R^{cd}_{ab} non-zero only for {c,d} = {a,b}	True
3	G^0_0 = 3Ha^2 + 12 Ha Hb + 6 Hb^2	True
4	G^1_1 = 2Ha' + 3Ha^2 + 4Hb' + 10Hb^2 + 8HaHb	True
5	G^4_4 = 3Ha' + 6Ha^2 + 3Hb' + 6Hb^2 + 9HaHb	True
6	scalar curvature R = -2(3Ha' + 6Ha^2 + 4Hb' + 10Hb^2 + 12 HaHb)	True
7	E^(1) = Einstein tensor G (cosmological metric)	True
8	L_1 = R	True
9	E^(0) = -delta/2	True
10	trace identity E^(k)mu_mu = -(8-2k)/2 L_k, k = 0..4	True
11	E^(4) = 0 identically in 8 dimensions, but L_4 (Euler density) is not 0	True
12	E^(k) diagonal: E^(k)mu_nu = 0 for mu != nu, and E^(k)1_1 = E^(k)2_2 = E^(k)3_3, E^(k)4_4 = ... = E^(k)7_7	True
13	E^(2)0_0 = -12 Hb (4Ha^3 + 18Ha^2Hb + 12HaHb^2 + Hb^3)	True
14	E^(3)0_0 = 360 Ha^2 Hb^3 (4Ha + 3Hb)	True
15	E^(2) = Lanczos tensor 2(R R_mn - 2R_ma R^a_n - 2R^ab R_manb + R_m^abc R_nabc) - g_mn L_2/2, and L_2 = R^2 - 4 Ric.Ric + Riem.Riem (cosmological metric)	True
16	divergence-free E^(k), k = 1, 2, 3 (cosmological metric, arbitrary a, b)	True
17	variation of sqrt|g| L_k (lapse N, a, b): EL_N = -2 a^3 b^4 E^(k)0_0, EL_a = -6 a^2 b^4 E^(k)1_1, EL_b = -8 a^3 b^3 E^(k)4_4, k = 1..4	True
18	inhomogeneous metric (N, A, B of x0 and x4): E^(1) = G at the point	True
19	inhomogeneous metric: nabla_mu E^(k)mu_nu = 0 exactly at x0 = 1/2, x4 = 1/3, k = 1, 2, 3	True
20	inhomogeneous metric: E^(k)_mu nu symmetric, k = 1, 2, 3	True
21	pp-wave: E^(1)_mu nu = +(1/2) (d2^2 + d3^2) F l_mu l_nu, l = dx0 - dx1	True
22	pp-wave: E^(2) = E^(3) = 0 for every profile F	True
23	Einstein Kasner: p = (1+-2 sqrt2)/7, q = (2-+3 sqrt2)/14 solve E^(1) = 0; 3p+4q = 1, 3p^2+4q^2 = 1	True
24	Gauss-Bonnet Kasner: the three (p, (3-3p)/4) solve E^(2) = 0, sum of the 7 exponents = 3	True
25	cubic Lovelock Kasner: (p, q) = (-15/7, 20/7) solves E^(3) = 0, 3p + 4q = 5; degenerate families q = 0 and p = 0	True
26	Einstein Kasner is not a cubic-Lovelock vacuum (E^(3) != 0)	True
27	8D de Sitter (a = b = e^{H x0}): E^(1) = 21 H^2, E^(2) = -420 H^4, E^(3) = 2520 H^6 (times delta)	True
28	4D de Sitter on a static T^4 (a = e^{H x0}, b = b0): E^(1) = diag(3,3,6) H^2, E^(2) = diag(0,0,-12) H^4, E^(3) = 0	True
29	static T^4 (Hb = 0): E^(1) = (3Ha^2, 2Ha'+3Ha^2, 3Ha'+6Ha^2), E^(2) = (0, 0, -12 Ha^2 (Ha'+Ha^2)), E^(3) = 0	True
30	exponential expansion a = e^{H x0}, b = e^{q H x0}: E^(k) (00, 11, 44) = closed forms with Ha = H, Hb = q H, Ha' = Hb' = 0 (k = 1, 2, 3)	True
31	exponential expansion: 3 Ha (eq00 - eq11) + 4 Hb (eq00 - eq44) = 0 identically (the three vacuum equations have rank 2)	True
32	exponential vacuum solutions: one solution (w2, w3) for given w1, Lambda, q, satisfying all three equations	True
33	eternal inflation a = e^{H x0} with deflation b = e^{-H x0}: w2 = Lambda/4 - w1/2, w3 = w1/24 - Lambda/36 (Lambda = 0: w2/w1 = -1/2, w3/w1 = 1/24)	True
34	cosmological metric: eta e e = g	True
35	cosmological metric: {Gamma_mu, Gamma_nu} = 2 g_mu nu	True
36	cosmological metric: Gamma_mu real	True
37	cosmological metric: omega_mu AB = -omega_mu BA	True
38	cosmological metric: vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0	True
39	cosmological metric: Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu	True
40	cosmological metric: total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0	True
41	cosmological metric: constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0	True
42	cosmological metric: spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B	True
43	cosmological metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0	True
44	cosmological metric: (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16	True
45	cosmological metric: Gamma^mu Omega_mu = (3a'/(2a) + 2b'/b) Gamma17[0]	True
46	cosmological metric: Omega_i = (a'/2) Gamma17[i] Gamma17[0] (i = 1..3), Omega_k = (b'/2) Gamma17[k] Gamma17[0] (k = 4..7), Omega_0 = 0	True
47	pp-wave: eta e e = g	True
48	pp-wave: {Gamma_mu, Gamma_nu} = 2 g_mu nu	True
49	pp-wave: Gamma_mu real	True
50	pp-wave: omega_mu AB = -omega_mu BA	True

#	check	result
51	pp-wave: vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0	True
52	pp-wave: Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu	True
53	pp-wave: total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0	True
54	pp-wave: constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0	True
55	pp-wave: spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B	True
56	pp-wave: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0	True
57	pp-wave: (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16	True
58	inhomogeneous metric N, A, B (x0, x4): eta e e = g	True
59	inhomogeneous metric N, A, B (x0, x4): {Gamma_mu, Gamma_nu} = 2 g_mu nu	True
60	inhomogeneous metric N, A, B (x0, x4): Gamma_mu real	True
61	inhomogeneous metric N, A, B (x0, x4): omega_mu AB = -omega_mu BA	True
62	inhomogeneous metric N, A, B (x0, x4): vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0	True
63	inhomogeneous metric N, A, B (x0, x4): Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu	True
64	inhomogeneous metric N, A, B (x0, x4): total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0	True
65	inhomogeneous metric N, A, B (x0, x4): constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0	True
66	inhomogeneous metric N, A, B (x0, x4): spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B	True
67	inhomogeneous metric N, A, B (x0, x4): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0	True
68	inhomogeneous metric N, A, B (x0, x4): (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16	True
69	Cartan's first structure equation de^A + omega^A_B ^ e^B = 0 (torsion-free) for the cosmological frame	True
70	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): eta e e = g	True
71	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): {Gamma_mu, Gamma_nu} = 2 g_mu nu	True
72	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Gamma_mu real	True
73	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): omega_mu AB = -omega_mu BA	True
74	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0	True
75	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu	True
76	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0	True
77	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0	True
78	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B	True
79	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0	True
80	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16	True
81	Cartan's first structure equation for the static radial metric	True
82	structure coefficients: de^A = (1/2) C^A_BC e^B ^ e^C and omega_C,AB (form index C) = -(1/2)(C_ACB + C_BAC - C_CBA) (static radial metric)	True
83	static radial metric: Omega_0 = (f'/(4 sqrt(f h))) Gamma17[1] Gamma17[0], Omega_1 = 0, Omega_k = (1/(2 sqrt h)) Gamma17[1] Gamma17[k] (k = 2..7)	True
84	static radial metric: Gamma^mu Omega_mu = (1/sqrt h)(f'/(4f) + 3/r) Gamma17up[1], so (i Gamma^mu D_mu - m) psi = [i Gamma17up[0] d0/sqrt f + i Gamma17up[1] (d_r + f'/(4f) + 3/r)/sqrt h + sum_k i Gamma17up[k] d_k/r - m] psi	True
85	static radial metric: psi = f^(-1/4) r^(-3) chi removes the spin connection: Gamma^mu D_mu psi = f^(-1/4) r^(-3) Gamma^mu d_mu chi	True
86	U(1) gauge covariance: Gamma^mu (D_mu - i q A'_mu)(e^{i q alpha} psi) = e^{i q alpha} Gamma^mu (D_mu - i q A_mu) psi with A' = A + d alpha	True
87	current identity with the gauge field: d_mu(sqrt|g| psibar Gamma^mu psi) = sqrt|g| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi], D including -i q A_mu	True
88	current identity d_mu(sqrt|g| psibar Gamma^mu psi) = sqrt|g| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi]	True
89	Lichnerowicz: (Gamma^mu D_mu)^2 psi = (g^mu nu nabla_mu D_nu - R/4) psi	True
90	rescaling psi = (a^3 b^4)^(-1/2) chi removes the spin connection: Gamma^mu D_mu psi = (a^3 b^4)^(-1/2) Gamma^mu d_mu chi	True

#	check	result
91	mode equation: (i Gamma^mu D_mu - m) psi = 0 <=> i dc/dx0 = h(x0) c, h = alpha.(p/a, q/b) + beta m	True
92	conformal weight 7/2: Gamma^mu D_mu (C^(-7/2) f) = C^(-9/2) Gamma^A d_A f for g = C(x0)^2 eta	True
93	2x2 reduction: A = alpha.n_a, B = alpha.n_b, beta: squares 1, mutually anticommuting; Z = A B beta: Z^2 = -1, Z^T = -Z, commutes with A, B, beta	True
94	Kasner: |alpha|^2 + |beta|^2 = 1 (to 1e-8)	True
95	Kasner scaling: (|p|, |q|) = (1.430..., 2.5) has K = 1 and the same |beta|^2 as (1, 1) (to 1e-6)	True
96	16x16 evolution = 4 copies of the 2x2 evolution with +m sigma3 and 4 copies with -m sigma3: average agrees (to 1e-8), unitarity; the two sectors differ	True
97	isotropic expansion, massless: no pair creation (|beta|^2 < 1e-20)	True
98	de Sitter: |beta|^2 = 1/(exp(2 pi m/H) + 1) (Fermi-Dirac at the Gibbons-Hawking temperature H/2pi), to 1e-7	True
99	integrator benchmark: Landau-Zener |beta|^2 = exp(-pi kappa^2) (to 1e-6)	True

9. How to reproduce

1. `python3 tools/make_7p1_nb.py` writes 7+1.nb (unevaluated) from tools/make_7p1_nb.py and tools/nb_part_*.py.
2. Evaluate 7+1.nb in a fresh kernel of Wolfram 15.0.1 (Evaluation ▷ Quit Kernel, then Evaluation ▷ Evaluate Notebook; or `Get["D:/Developer/github/7+1/tools/run_eval.wl"]`) and save it; the last section summarizes all checks.
3. `python3 tools/ref_gravity.py` repeats the checks of this document independently (exit code 0 if all pass).
4. `python3 tools/make_docs.py` regenerates this document, its symbol dictionary and its provenance record.